



# Designing Public Messages to Prevent Lottery Addiction

## : *An Experiment Study to Correct Cognitive Errors*



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# Why Should We Care About Lottery Addiction?

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- **Popularity and Scale:** Lottery is a popular leisure activity in Korea, with 2023 sales reaching a record 6.7507 trillion KRW, and Lotto accounting for 83.7% of this total (BBS News, June 4, 2024).
- **Harmful Effects:** Despite its lower risk compared to other gambling forms, excessive lottery activity can be harmful. Men, the elderly, and low-income groups are particularly vulnerable (Costes et al., 2018), and it can serve as a gateway to other risky behaviors (Subramaniam et al., 2016).
- ➡ Thus, the Korean government have strictly regulates the lottery industry (e.g., limiting individual purchases to 100,000 won, capping maximum prize amounts, allowing only cash purchases, prohibiting sales to teenagers, and mandating warning signs in promotions and advertisements) and has attempted to achieve these goals through public information campaigns.
- **Study Purpose:** The purpose of this study is to tackle this task by conducting an experiment for message testing based on a well-established theoretical framework. In particular, we focus on people's cognitive errors that have been found to affect their gambling behaviors (Fortune & Goodie, 2012).



# Social Judgment and Probability in Gambling

- People make social judgments about the likelihood of events with insufficient information.
  - In gambling, these judgments often lead to erroneous probability assessments.
  - Example: "I believe there is a heightened probability the ball of this roulette wheel will land in a black slot." (Fortune & Goodie, 2012, p.298)
- ➔ In this way, gamblers tend to make judgment errors due to incorrect perceptions of randomness (Lee, 2003), and these errors contribute to the continuation of gambling despite its negative outcomes (Jacobsen et al., 2007).





# Cognitive Distortions in Lottery Behavior

- Obsessive lottery behavior involves cognitive distortions similar to gambling.
- Common cognitive distortions:
  - Belief in special systems or skills for picking winning numbers.
  - Belief that dreams or prediction services increase chances of winning.
  - Belief that buying tickets at lucky stores increases winning odds. (i.e., "lucky store effect (Guryan & Kearney, 2008)")
- Consequences of these distortions:
  - Long queues at "lucky" stores with multiple winners.
  - Creation of lottery store conglomerates.
  - Companies selling "winning" numbers without scientific basis.



# Research Questions

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- Taken together, such cognitive distortions related to probability lead people who purchase lottery tickets, to have unrealistic expectations of winning the lottery, which can result in obsessive lottery behavior.
- Therefore, our study explored **effective messages strategies for correcting judgment errors** that lead to obsessive lottery activities.
- **We explore the following research questions:**
  - RQ1. Will providing a message to correct people's judgment errors be effective in preventing obsessive lottery behavior?
  - RQ2. Will people's cognitive distortions in lottery behavior moderate the message effectiveness?

# Experiment design

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- A three-group design with balanced randomization across three experimental conditions: probability correction messages, public service messages, and weather information messages (i.e., control group).
  - Number of words and their readability were considered.
  - Participants were randomly assigned to one of three experimental groups and exposed to a message.



# Message stimulus

| Condition                                       | Message exposed                                                                                                                              | <i>n</i> |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Probability Correction Messages                 | 구매한 로또 복권이 1등으로 당첨될 확률은 814만분의 1입니다. 특정 판매점이나 지역에서 로또 당첨자가 많다는 주장도 있습니다. 그러나 그 이유는 더 많은 사람이 복권을 구입하기 때문에 당첨 확률이 높아 보이는 것이지 당첨 확률은 모두가 똑같습니다. | 683      |
| Public Service Messages                         | 복권은 사회 발전에 공헌합니다. 복권 구입 금액의 42% 이상이 복권기금으로 조성되어 소외된 이웃을 위해 다양한 공익사업에 쓰이고 있습니다. 우리 주변 이웃들을 위해 쓰이는 복권 기금은 한 해 약 3조 원에 육박합니다. 복권의 다른 이름, 사랑입니다. | 678      |
| Weather Information Messages<br>(Control Group) | 1월 셋째 주의 날씨는 이동성 고기압과 대륙고기압의 영향을 받겠으나, 저기압의 영향을 받을 때가 있겠습니다. 주 평균기온은 평년과 비슷하거나 높을 확률이 각각 40%입니다. 주 강수량은 평년과 비슷하거나 많을 확률이 각각 40%입니다.          | 636      |



# Participants

- An online survey with three experiment message conditions was conducted among Koreans aged 14 to 69 from January 26 to 31, 2024. Among 3,000 collected cases, two respondent groups excluded from the final analysis were teenagers and people who bought no lottery tickets in the previous year ( $n = 1,997$ ).
- Within the final sample, 58.09% were female ( $n = 1,160$ ), with a mean age of 44.07 years ( $SD = 13.37$ ).

Table 1. Characteristics of respondents.

| Variable           |                                      | <i>n</i> (%)  |
|--------------------|--------------------------------------|---------------|
| Gender             | Male                                 | 837 (41.91)   |
|                    | Female                               | 1,160 (58.09) |
| Education          | Under Junior high school graduate    | 11 (0.55)     |
|                    | High School graduate                 | 301 (15.07)   |
|                    | Vocational college enrolled/graduate | 289 (14.47)   |
|                    | College graduate                     | 1,178 (58.98) |
|                    | Graduate school or higher            | 218 (10.91)   |
| Income (household) | 2 million won or less                | 138 (6.91)    |
|                    | Less than 2 to 4 million won         | 565 (28.29)   |
|                    | Less than 4 to 6 million won         | 563 (28.19)   |
|                    | Less than 6 to 8 million won         | 368 (18.42)   |
|                    | Less than 8 to 10 million won        | 202 (10.11)   |
|                    | More than 10 million won             | 161 (8.06)    |

# Measures

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### Dependent variables

#### **(1) Perceived winning probability** ( $M = 3.3$ , $SD = 0.98$ )

- The perceived winning probability was assessed using a single-item measure, "Lottery winning probability is the same no matter where it is purchased" (5-point Likert scale: 1= "strongly disagree" , 5= "strongly agree").

#### **(2) Intention for obsessive lottery behavior** (5-point Likert scale: 1= "strongly disagree" , 5= "strongly agree")

- "I plan to buy lottery tickets from a store where many winning tickets have been sold."
- "I plan to buy lottery tickets from a store where many winning tickets have been sold, even if I have to stand in line."
- "I plan to purchase a lottery number prediction service."
- (EFA = 56 % of total variance explained; Cronbach's alpha = .72 [95% CI: .7, .74]); the composite score was constructed using the average of three items ( $M = 2.7$ ,  $SD = 0.8$ )

### Moderator

**Cognitive distortion** (5-point Likert scale: 1= "strongly disagree" , 5= "strongly agree").

- Cognitive distortions generally reflect the belief that gambling outcomes can be predicted and the belief that they can be controlled (McInnes, Hodgins, & Holy, 2014).
- Items were drawn from the Korean version of the Gambling Belief Questionnaire (GBQ) (Back et al., 2019), a self-report measure that was designed to identify gambling-related cognitive distortions (Steenbergh, Meyers, May, & Whelan, 2002).
- As a result of conducting an EFA on the seven items, a total of three factors were identified (accounts for 57% of the total variability). We designated the factors as 'Illusion of control', 'Predictive control', and 'Superstitions'.



# Measures

## Moderator

**Cognitive distortion** (5-point Likert scale: 1= "strongly disagree" , 5= "strongly agree").

| Variables                                         | Items                                                                                                                             |
|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Illusion of control<br>( $r = .71$ )              | I have a special system for selecting lottery numbers.                                                                            |
|                                                   | I have a lucky technique for selecting lottery numbers.                                                                           |
| Predictive control<br>( $\alpha = .7$ [.68, .73]) | I mainly refer to the numbers chosen by previous winners when selecting my lottery numbers.                                       |
|                                                   | Using a lottery number prediction service can create winning tickets.                                                             |
| Superstitions<br>( $r = .54$ )                    | Manually selecting lottery numbers increases the chances of winning compared to having them automatically chosen by the terminal. |
|                                                   | Having good dreams increases the chances of winning the lottery.                                                                  |
|                                                   | Buying lottery tickets from a store where many winning tickets have been sold increases the chances of winning.                   |

# Analytic strategy

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- First, to examine whether there were differences in individuals' perceived winning probability and intention for obsessive lottery behaviors according to the messages presented in the experiment (RQ1), we conducted a one-way ANOVA.
- Second, to investigate whether people's cognitive distortions in lottery behavior moderate the effectiveness of the messages (RQ2), we utilized the PROCESS model (model 1) in JASP (JASP Team, 2024).

# Descriptive statistics and correlations

Table 2. Descriptive statistics and correlations.

| Variable                                     | <i>M</i> | <i>SD</i> | 1      | 2       | 3       | 4       |
|----------------------------------------------|----------|-----------|--------|---------|---------|---------|
| 1. Illusion of Control                       | 1.86     | 0.80      |        |         |         |         |
| 2. Predictive Control                        | 2.15     | 0.74      | .60*** |         |         |         |
| 3. Superstitions                             | 2.72     | 0.96      | .33*** | .51***  |         |         |
| 4. Perceived Winning Probability             | 3.30     | 0.98      | 0.00   | -.08*** | -.22*** |         |
| 5. Intention for Obsessive Lottery Behaviors | 2.70     | 0.80      | .33*** | .48***  | .47***  | -.28*** |



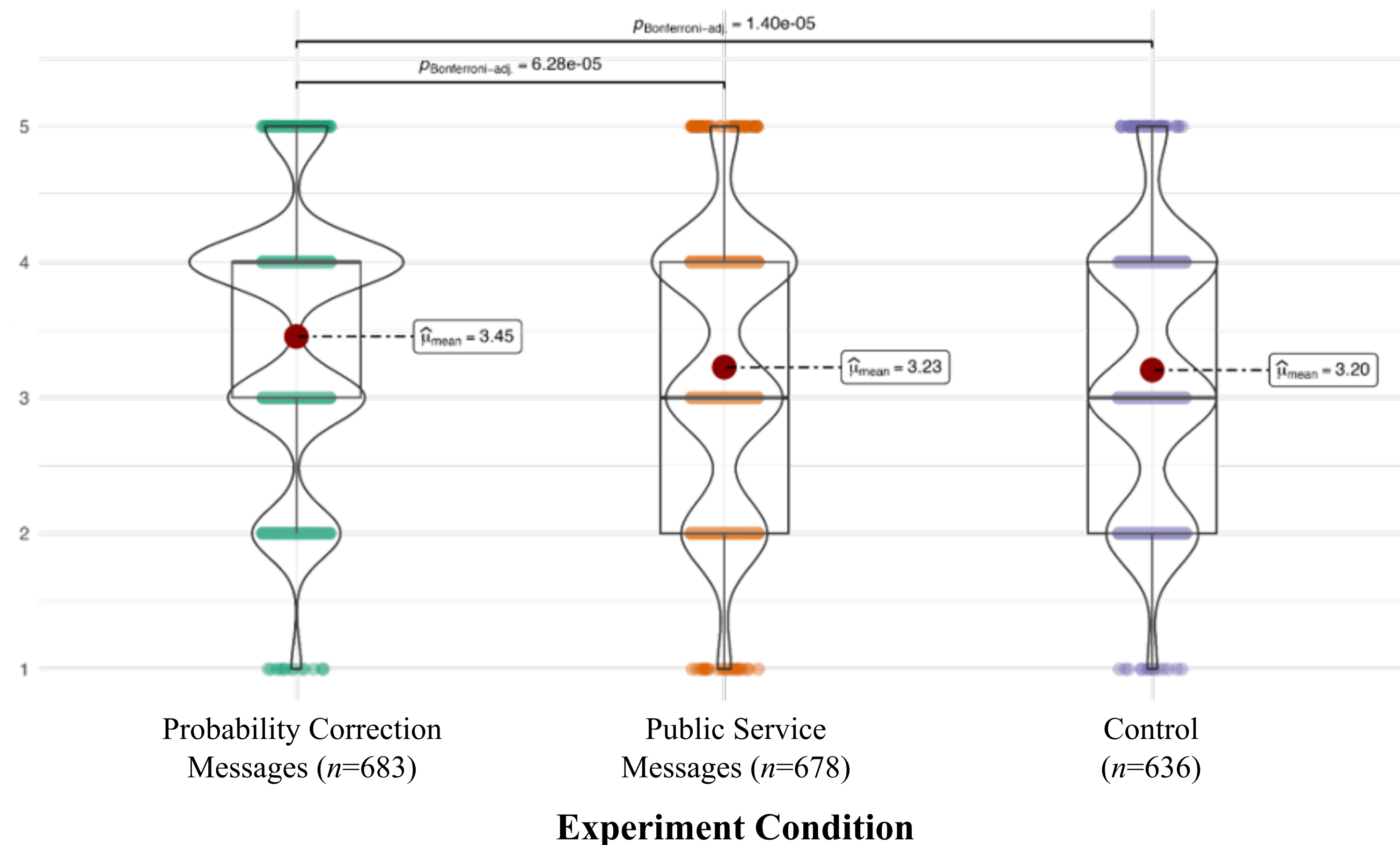
# RQ1. Message effect

- Post-hoc analyses with Bonferroni correction revealed that the *probability correction message* group ( $M=3.45$ ,  $SD=0.98$ ) exhibited a more corrected winning probability compared to the *public service message* group ( $M=3.23$ ,  $SD=0.98$ ) and *control* group ( $M=3.20$ ,  $SD=0.97$ ).

➔ Thus, it seems that providing a message to correct leads to a **positive effect in instilling individuals with accurate perceptions** of the probability associated with lottery winnings.

## Comparison of Perceived Winning Probability Across Experiment Condition

$$F_{\text{Fisher}}(2, 1994) = 13.21, p = 1.99\text{e-}06, \hat{\omega}_p^2 = 0.01, \text{CI}_{95\%} [4.99\text{e-}03, 1.00], n_{\text{obs}} = 1,997$$



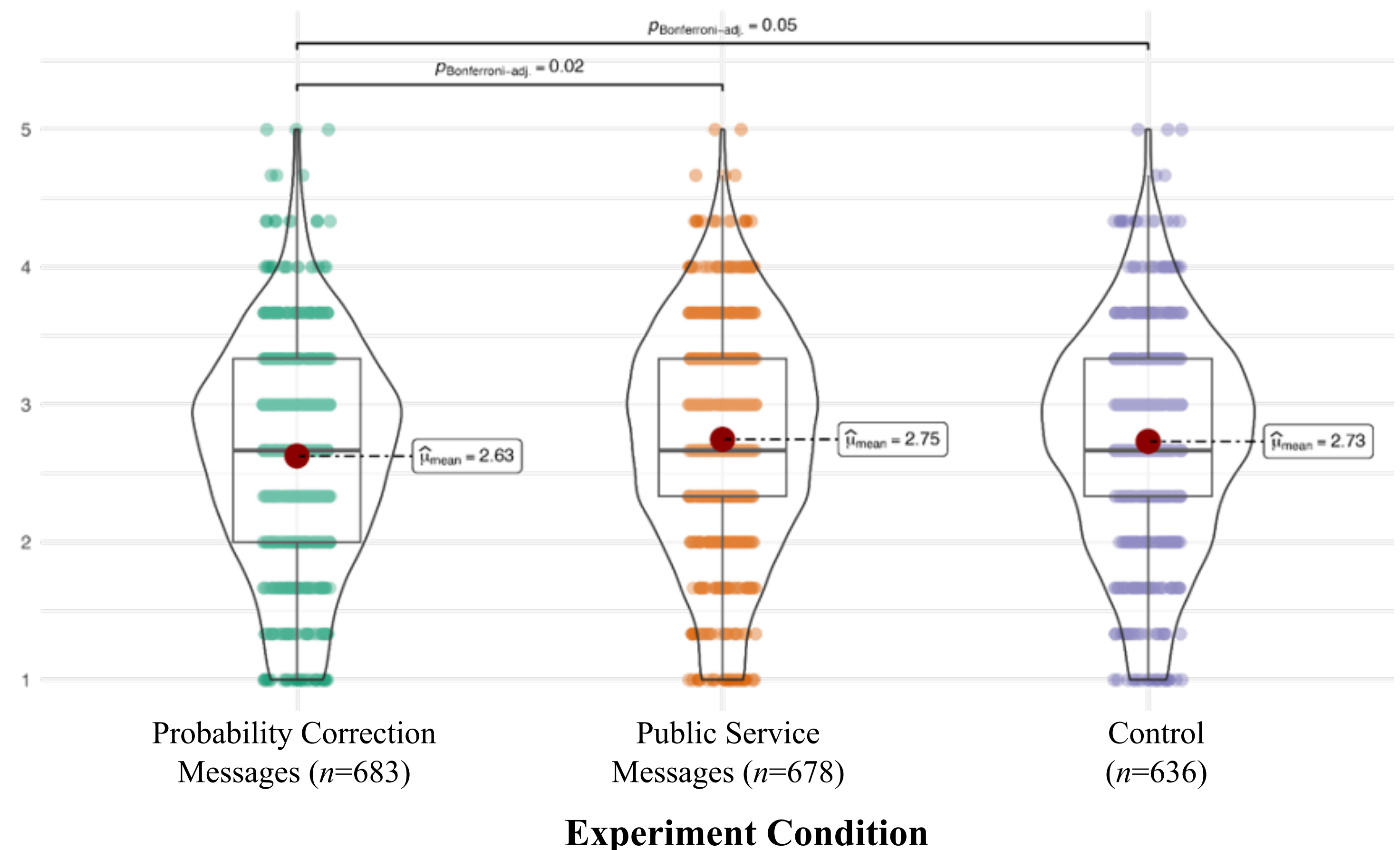
# RQ1. Message effect

- Post-hoc analyses with Bonferroni correction revealed that the *probability correction message* group (M=2.63, SD=0.8) exhibited a lower intention to engage in faulty lottery behaviors compared to the *public service message* group (M=2.75, SD=0.81) and *control* group (M=2.73, SD=0.79).

➔ Thus, it seems that providing a corrective message appears to have a **positive effect on reducing individuals' intentions for obsessive lottery behaviors** stemming from cognitive errors regarding lottery winnings.

## Comparison of Intention for Obsessive Lottery Behaviors Across Experiment Condition

$$F_{\text{Fisher}}(2, 1994) = 4.63, p = 9.84\text{e-}03, \hat{\omega}_p^2 = 3.62\text{e-}03, \text{CI}_{95\%} [2.22\text{e-}04, 1.00], n_{\text{obs}} = 1,997$$



RQ2. The moderating role of cognitive distortion on the message effect<sup>17</sup>

- Based on moderation analysis, cognitive distortions (i.e., Illusion of control, Predictive control, Superstitions) **were found not to moderate the effects of messages** on both **accurate perceptions of the probability associated with lottery winnings** and intentions for obsessive lottery behaviors.

Table 3. Interaction effect of message condition and cognitive distortion, predicting perceived winning probability (*n*=1,997)

| Interaction                                       | <i>beta</i>            | <i>SE</i> | <i>p</i> | LLCI   | ULCI  |
|---------------------------------------------------|------------------------|-----------|----------|--------|-------|
| Model A: Moderator variable = Illusion of control |                        |           |          |        |       |
| ConditionB:Illusion of control                    | 0.003                  | 0.065     | 0.967    | -0.126 | 0.131 |
| ConditionC:Illusion of control                    | 0.005                  | 0.067     | 0.939    | -0.125 | 0.135 |
| Model B: Moderator variable = Predictive control  |                        |           |          |        |       |
| ConditionB:Predictive control                     | 3.097×10 <sup>-5</sup> | 0.070     | 1.000    | -0.137 | 0.137 |
| ConditionC:Predictive control                     | -0.056                 | 0.074     | 0.451    | -0.200 | 0.089 |
| Model C: Moderator variable = Superstitions       |                        |           |          |        |       |
| ConditionB:Superstitions                          | -0.034                 | 0.053     | 0.526    | -0.139 | 0.071 |
| ConditionC:Superstitions                          | -0.046                 | 0.054     | 0.394    | -0.152 | 0.060 |



RQ2. The moderating role of cognitive distortion on the message effect<sup>18</sup>

- Based on moderation analysis, cognitive distortions (i.e., Illusion of control, Predictive control, Superstitions) **were found not to moderate the effects of messages** on both accurate perceptions of the probability associated with lottery winnings and **intentions to engage in faulty lottery behaviors**.

Table 4. Interaction effect of message condition and cognitive distortion, predicting intention for obsessive lottery behaviors (*n*=1,997)

| Interaction                                       | <i>beta</i> | <i>SE</i> | <i>p</i> | LLCI   | ULCI  |
|---------------------------------------------------|-------------|-----------|----------|--------|-------|
| Model D: Moderator variable = Illusion of control |             |           |          |        |       |
| ConditionB:Illusion of control                    | 0.097       | 0.051     | 0.055    | -0.002 | 0.196 |
| ConditionC:Illusion of control                    | -0.060      | 0.051     | 0.243    | -0.161 | 0.041 |
| Model E: Moderator variable = Predictive control  |             |           |          |        |       |
| ConditionB:Predictive control                     | -0.002      | 0.050     | 0.967    | -0.101 | 0.097 |
| ConditionC:Predictive control                     | 0.011       | 0.053     | 0.835    | -0.093 | 0.115 |
| Model F: Moderator variable = Superstitions       |             |           |          |        |       |
| ConditionB:Superstitions                          | -0.005      | 0.040     | 0.895    | -0.083 | 0.072 |
| ConditionC:Superstitions                          | -0.008      | 0.040     | 0.838    | -0.087 | 0.071 |



RQ2. The moderating role of cognitive distortion on the message effect<sup>19</sup>

- Rather, these **cognitive distortions exerted direct effects** on both perceived winning probability (Model A~C) and intention for obsessive lottery behaviors (Model D~F).

Table 5. Main effect of cognitive distortion, predicting perceived winning probability and intention for obsessive lottery behaviors (*n*=1,997)

| Main effect                   | <i>beta</i>            | <i>SE</i> | <i>p</i> | LLCI   | ULCI   |
|-------------------------------|------------------------|-----------|----------|--------|--------|
| (Model A) Illusion of control | 6.065×10 <sup>-4</sup> | 0.045     | 0.989    | -0.088 | 0.089  |
| (Model B) Predictive control  | -0.089                 | 0.049     | 0.071    | -0.186 | 0.008  |
| (Model C) Superstitions       | -0.200                 | 0.038     | < .001   | -0.274 | -0.126 |
| (Model D) Illusion of control | 0.316                  | 0.035     | < .001   | 0.248  | 0.385  |
| (Model E) Predictive control  | 0.520                  | 0.036     | < .001   | 0.450  | 0.590  |
| (Model F) Superstitions       | 0.398                  | 0.028     | < .001   | 0.343  | 0.453  |

# Conclusion

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- **Our findings** highlight the importance of informing people of accurate winning probability of lottery through public information campaigns. Engaging and attention-getting communication strategies are needed to address cognitive distortions about lottery probabilities and to promote responsible lottery decision-making.
  - e.g., "A warning statement on the probability of winning lottery should be printed on the lottery tickets; not simply a statement on one in 14 million probability but a statement that may be more readily understood by laymen such as 'if you bought a ticket every day from birth and lived to be 100 it would take you 383 life times to win a lotto.'  
(Ariyabuddhiphong, 2011, p.27)
- **Limitations:** only one experiment message (lacking message variability) and incompletely controlled study settings (i.e., experiment through an online survey)
- **Future research** could (a) explore additional factors that may mediate or moderate the effectiveness of corrective messaging and (b) further investigate the mechanisms underlying the relationship between cognitive distortions and lottery-related behaviors.

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